Astrodynamics I: Final Exam

June 13, 2012, 9:00-10:45AM

1. (20) At the end of a space vehicle launch from a launch site on the Earth's equator, the following burnout conditions hold: the burnout altitude is 120 km; the burnout velocity vector lies in the equatorial plane with magnitude 8.2 km/sec and with flight path angle of 2 degrees above the local horizontal. After burnout, the space vehicle moves along some orbit. (Use $R_E = 6{,}378km$ and $\mu_E = 398{,}600km^3/sec^2$, and $e = \sqrt{1 + \frac{2Eh^2}{\mu_E^2}}$.)
   a. (2) What is the specific angular momentum of the orbit of the space vehicle?
   b. (2) What is the specific energy of the orbit of the space vehicle?
   c. (2) What is the orbital eccentricity?
   d. (4) What is the equation of the conic section (trajectory equation) that describes the orbit? Instead of true anomaly, use the angle θ which is measured relative to the radius vector at burnout, with increasing θ taken in the direction along which the space vehicle moves.
   e. (4) What are the radius of perigee and the radius of apogee of the orbit?
   f. (4) What are the orbital velocities at perigee and at apogee?
   g. (2) What is the orbital period?

2. (40) For an Earth-orbiting satellite, a pair of position and velocity vectors $(\vec{r}_0, \vec{v}_0)$ is known at a specified time t_0 in the ECI coordinate system.
   a. (20) Describe how to obtain the classical orbital elements $(a, e, i, \Omega, \omega, \nu)$ for inclined (non-equatorial) elliptic orbits.
   b. (20) Describe the procedure of obtaining a pair of position and velocity vectors $(\vec{r}(t), \vec{v}(t))$ at any other time t. Hint: Use the relation $\tan\left(\frac{\nu}{2}\right) = \sqrt{\frac{1+e}{1-e}}\tan\left(\frac{E}{2}\right)$ if necessary.

3. (30) For an Earth-orbiting satellite, a pair of position and velocity vectors $(\vec{r}_{IJK}, \vec{v}_{IJK})$ in the ECI coordinate system and the site location is known by its location parameters $(\vec{r}_{SiteIJK}, \Phi_{gd}, \theta_{LST})$.
   a. (15) Derive the rotation matrix for transformation from ECI frame (IJK) to Topocentric frame (SEZ).
   b. (15) Describe the procedure of computing range (ρ), azimuth (β), elevation (el), and their time rates.

4. (10) State the definition of following terminologies PRECISELY:
   a. (3) Kepler's Problem
   b. (3) Satellite Coordinate System (denoted by RSW) = Gaussian Coordinate System
   c. (2) Universal Time
   d. (2) Equation of Equinoxes